

JOB APPLICATION TRACKING SYSTEM (MERN + Docker + AWS)

Developer: Vivek Goal: Interview Ready Project + AWS + Docker + CI/CD

===== TECH STACK
=====

Frontend

- React
- React Router
- Axios
- Redux Toolkit
- Bootstrap 5 or Tailwind

Backend

- Node.js
- Express.js
- JWT Authentication
- Bcrypt

Database

- MongoDB Atlas

DevOps

- Docker
- Docker Compose
- Git
- GitHub
- GitHub Actions

Cloud

- AWS EC2
- AWS S3
- AWS ECR (Optional)

===== PHASE 1 - PROJECT SETUP
=====

Estimated Time: 4 Hours

Tasks

☐ Create GitHub Repository ☐ Create Frontend React App ☐ Create Backend Node App ☐ Setup Express Server ☐ Connect MongoDB ☐ Create Environment Variables ☐ Setup Folder Structure

Deliverable

Frontend Running Backend Running MongoDB Connected

===== PHASE 2 - AUTHENTICATION
=====

Estimated Time: 1 Day

Pages

☐ Login ☐ Register ☐ Forgot Password ☐ Reset Password

Backend

☐ User Model ☐ JWT Middleware ☐ Password Encryption ☐ Authentication APIs

Deliverable

User Registration User Login Protected Routes

===== PHASE 3 - DASHBOARD
=====

Estimated Time: 1 Day

Pages

☐ Dashboard

Cards

☐ Total Applications ☐ Interviews ☐ Offers ☐ Rejections

Charts

☐ Applications Per Month ☐ Status Analytics

Deliverable

Dashboard Statistics Working

===== PHASE 4 - JOB MANAGEMENT
=====

Estimated Time: 2 Days

Pages

☐ Job Listing ☐ Add Job ☐ Edit Job ☐ View Job

Fields

☐ Company ☐ Position ☐ Location ☐ Salary ☐ Job URL ☐ Description ☐ Status

Status

Applied Interview Offer Rejected

Backend

☐ CRUD APIs

Deliverable

Complete Job Management

===== PHASE 5 - SEARCH & FILTER
=====

Estimated Time: 4 Hours

Features

☐ Search by Company ☐ Search by Position ☐ Filter by Status ☐ Pagination ☐ Sorting

Deliverable

Professional Data Listing

===== PHASE 6 - PROFILE MANAGEMENT
=====

Estimated Time: 4 Hours

Pages

☐ My Profile

Features

☐ Update Name ☐ Update Email ☐ Change Password ☐ Profile Picture

Deliverable

User Profile Working

===== PHASE 7 - RESUME MANAGEMENT
=====

Estimated Time: 1 Day

Pages

☐ Resume Upload

Features

☐ Upload Resume PDF ☐ View Resume ☐ Delete Resume

AWS

☐ S3 Bucket Integration

Deliverable

Resume Stored in AWS S3

===== PHASE 8 - NOTES MANAGEMENT
=====

Estimated Time: 4 Hours

Features

☐ Add Notes ☐ Edit Notes ☐ Delete Notes

Examples

Interview Questions HR Discussion Salary Discussion

Deliverable

Job Specific Notes

===== PHASE 9 - ADMIN PANEL
=====

Estimated Time: 1 Day

Pages

☐ Admin Dashboard ☐ User Management

Features

☐ View Users ☐ Delete Users ☐ Statistics

Roles

Admin User

Deliverable

Role Based Access Control

===== PHASE 10 - DOCKER
=====

Estimated Time: 1 Day

Tasks

☐ Frontend Dockerfile ☐ Backend Dockerfile ☐ Docker Compose ☐ Environment Variables

Commands

`docker compose up -d`

Deliverable

Entire Project Running In Docker

===== PHASE 11 - AWS DEPLOYMENT
=====

Estimated Time: 1 Day

Services

☐ EC2 ☐ Nginx ☐ SSL

Deployment

GitHub ↓ EC2 ↓ Docker Containers

Deliverable

Live Project URL

===== PHASE 12 - CI/CD
=====

Estimated Time: 1 Day

GitHub Actions

Pipeline

Push Code ↓ Build Docker ↓ Deploy Server

Deliverable

Automatic Deployment

===== BONUS FEATURES
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☐ Dark Mode ☐ Email Notifications ☐ Interview Reminder ☐ Export CSV ☐ Export PDF ☐ Mobile Responsive Design ☐ Swagger Documentation

===== INTERVIEW QUESTIONS COVERED
=====

React Redux NodeJS ExpressJS MongoDB JWT REST APIs Docker AWS Git GitHub Actions CI/CD
Authentication Authorization File Upload Cloud Storage

===== TOTAL TIME
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Phase 1-8 : 8-10 Days Docker : 1 Day AWS : 1 Day CI/CD : 1 Day Testing : 2 Days

Total Interview Ready Version

12-15 Days

===== END GOAL
=====

Resume Project:

"Built a full-stack Job Application Tracking System using MERN Stack, Docker, AWS EC2, AWS S3, MongoDB Atlas and GitHub Actions CI/CD pipeline with JWT authentication, role-based authorization and analytics dashboard."